Time Frequency Analysis and Wavelet Transforms
Homework 2

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1. **Advantages:**

   Since each $X(n\Delta t, m\Delta f)$ is obtained from $X((n-1)\Delta t, m\Delta f)$ by removing one sample and adding one sample, the recursive method has the lowest complexity, i.e., $O(TF)$, where T and F are the numbers of sampling points on the t- and f-axes.

   Moreover, only the required F frequency points are computed, while the FFT-based method always computes all N points.

   Disadvantages:

   It works only for the rec-STFT, and the constraints of the FFT-based method still hold, i.e., $\Delta t \Delta f$ must be the reciprocal of an integer N with $N \geq 2Q+1$, and $\Delta t < 1/(2\Omega)$.

   Also, the round-off error accumulates along the t-axis.

2. Since $x(t) = \cos(6\pi t + 1)$ is real, $x^*(t) = x(t)$ and $x\big(t \pm \frac{\tau}{2}\big) = \cos(6\pi t \pm 3\pi\tau + 1)$.

   Using $\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$ and $\int e^{-j2\pi(f-f_0)\tau} d\tau = \delta(f - f_0)$.

   We have

$$\begin{aligned}
W_x(t,f) &= \int_{-\infty}^{\infty} x\big(t+\tfrac{\tau}{2}\big)\, x^*\big(t-\tfrac{\tau}{2}\big)\, e^{-j2\pi f\tau}\, d\tau \\
&= \int_{-\infty}^{\infty} \cos(6\pi t + 3\pi\tau + 1)\cos(6\pi t - 3\pi\tau + 1)\, e^{-j2\pi f\tau} d\tau \\
&= \int_{-\infty}^{\infty} \Big[\tfrac{1}{2}\cos(6\pi\tau) + \tfrac{1}{2}\cos(12\pi t + 2)\Big] e^{-j2\pi f\tau} d\tau \\
&= \int_{-\infty}^{\infty} \Big[\tfrac{1}{4}e^{j6\pi\tau} + \tfrac{1}{4}e^{-j6\pi\tau} + \tfrac{1}{2}\cos(12\pi t + 2)\Big] e^{-j2\pi f\tau} d\tau \\
&= \tfrac{1}{4}\delta(f-3) + \tfrac{1}{4}\delta(f+3) + \tfrac{1}{2}\cos(12\pi t + 2)\,\delta(f)
\end{aligned}$$

3. The WDF has cross terms when the signal has more than one component, or the phase has an order larger than 2.

   With these two conditions , we check each case:

(a) $\delta(-\pi t)$: **No.** $\delta(-\pi t) = \frac{1}{\pi}\delta(t)$ is a single component, and $W_x(t,f) = \frac{1}{\pi^2}\delta(t)$.

(b) $\exp(-\pi t^4)$: **No.** It is a single real pulse with zero phase, so its WDF is a single lobe around $(t,f) = (0,0)$.

(c) $\exp(j\pi t^4)$: **Yes.** The phase has order 4:

$$x(t+\tfrac{\tau}{2})x^*(t-\tfrac{\tau}{2}) = e^{j2\pi(2t^3)\tau}\, e^{j\pi t\tau^3}.$$

The extra factor $e^{j\pi t\tau^3}$ spreads $\delta(f-2t^3)$ into oscillating interference.

(d) $\cos(-\pi t^2)$: **Yes.** $\cos(\pi t^2) = \frac{1}{2}\left[e^{j\pi t^2} + e^{-j\pi t^2}\right]$ has two components, so

$$W_x(t,f) = \tfrac{1}{4}\big[\delta(f-t) + \delta(f+t)\big] + \underbrace{\tfrac{\sqrt{2}}{2}\cos\big(2\pi(t^2-f^2) + \tfrac{\pi}{4}\big)}_{\text{cross term}}.$$

(e) A music signal: **Yes.** It contains many notes and harmonics, i.e., many components, and every pair of components produces cross terms.

Therefore, (c), (d), and (e) have the cross-term problem.

4. (a) **Cohen's class distribution.** With the ambiguity function (AF)

$$A_x(\tau,\eta) = \int x(t+\tfrac{\tau}{2})\, x^*(t-\tfrac{\tau}{2})\, e^{-j2\pi t\eta} dt,$$

the WDF is the 2-D FT of the AF: $W_x(t,f) = \iint A_x(\tau,\eta)\, e^{j2\pi(\eta t - \tau f)} d\eta\, d\tau$. Consider $x = x_1 + x_2$ with components centered at (t_1,f_1) and (t_2,f_2):

- The auto terms A_{x_1}, A_{x_2} are always located near the origin of the (τ,η) plane, since $|A_{x_k}(\tau,\eta)|$ depends only on the shape, while the position (t_k,f_k) only contributes a phase.
- The cross terms $A_{x_1x_2}$, $A_{x_2x_1}$ are located at $(\tau,\eta) = \pm(t_1-t_2,\ f_1-f_2)$, i.e., far from the origin when the components are separated.

The Cohen's class distribution

$$C_x(t,f) = \iint A_x(\tau,\eta)\, \Phi(\tau,\eta)\, e^{j2\pi(\eta t - \tau f)} d\eta\, d\tau$$

chooses $\Phi(\tau,\eta)$ as a low-pass function: $\Phi \approx 1$ near the origin, where the auto terms are, and $\Phi \approx 0$ in the region where the cross terms are. For example, the Choi-Williams kernel $\Phi(\tau,\eta) = e^{-\alpha(\eta\tau)^2}$ equals 1 on the τ- and η-axes and decays away from them. Since the cross terms at $\pm(t_1-t_2, f_1-f_2)$ are usually off the axes, they are suppressed. Thus, the auto terms are kept and the cross terms are filtered out before transforming back. ($\Phi = 1$ gives the WDF.) The cost is a tradeoff: a stronger low-pass filter removes more cross terms but also blurs the auto terms.

(b) **Polynomial WDF.** It replaces $x(t+\frac{\tau}{2})x^*(t-\frac{\tau}{2})$ by a product of more terms:

$$W_x(t,f) = \int \prod_{l=1}^{q/2} x(t+d_l\tau)\, x^*(t-d_{-l}\tau)\, e^{-j2\pi f\tau} d\tau.$$

For $x(t) = \exp\big(j2\pi \sum_{n=1}^{q/2+1} a_n t^n\big)$, d_l are chosen such that all terms with $\tau^2, \tau^3, \ldots$ cancel and

$$\prod_{l=1}^{q/2} x(t+d_l\tau)\, x^*(t-d_{-l}\tau) = \exp\Big(j2\pi\tau \sum_{n=1}^{q/2+1} n a_n t^{n-1}\Big),$$

which is linear in τ. Hence $W_x(t,f) = \delta\big(f - \sum_n n a_n t^{n-1}\big)$, exactly on the instantaneous frequency. Therefore, the polynomial WDF avoids the cross term when the order of the phase is no larger than $q/2+1$.

- For $q=2$, $d_1 = d_{-1} = \frac{1}{2}$ gives the original WDF, which can only handle a phase of order up to 2.
- For $q=4$, the conditions

  $$\textstyle\sum_l d_l + \sum_l d_{-l} = 1, \quad \sum_l d_l^2 - \sum_l d_{-l}^2 = 0, \quad \sum_l d_l^3 + \sum_l d_{-l}^3 = 0$$

  cancel the τ^2 and τ^3 terms, so a phase of order up to 3 can be handled.

Thus it avoids the interference of a single component with a higher-order phase. For example, Problem 3(c) $\exp(j\pi t^4)$ has a phase of order 4, so $q \geq 6$ is required. However, the cross terms between two different components cannot be removed.

(c) **Gabor-Wigner transform.** It combines the two distributions, e.g., $D_x(t,f) = G_x(t,f)\, W_x(t,f)$, $\min\big(|G_x(t,f)|^2, |W_x(t,f)|\big)$, or $G_x^\alpha(t,f)\, W_x^\beta(t,f)$. The Gabor transform is linear, $G_{x_1+x_2} = G_{x_1} + G_{x_2}$, so it has no cross term: $G_x \approx 0$ wherever there is no signal component, but its clarity is poor. The WDF has high clarity but cross terms. Multiplying them:

- at the cross terms of W_x, $G_x \approx 0$, so $D_x \approx 0$ and the cross terms are removed;
- at the auto terms, both are nonzero, and the product keeps the high clarity of the WDF.

For example, for $x(t) = \cos(2\pi t)$, the cross term of W_x at $f=0$ is removed since $G_x(t,0) \approx 0$.

5. (a) From large to small:

   (iv) Cohen's class > (ii) Gabor-Wigner > (iii) windowed WDF > (i) rec-STFT.

   (b) The windowed WDF is

   $$W_x^w(t,f) = \int w(\tau)\, x(t+\tfrac{\tau}{2})\, x^*(t-\tfrac{\tau}{2})\, e^{-j2\pi f\tau} d\tau.$$

Taking the conjugate and substituting $\tau = -u$:

$$\begin{aligned}\left[W_x^w(t,f)\right]^* &= \int w^*(\tau)\, x^*(t+\tfrac{\tau}{2})\, x(t-\tfrac{\tau}{2})\, e^{j2\pi f\tau} d\tau \\ &= \int w^*(-u)\, x(t+\tfrac{u}{2})\, x^*(t-\tfrac{u}{2})\, e^{-j2\pi fu} du.\end{aligned}$$

Comparing with $W_x^w(t,f)$, it is real for every $x(t)$ if and only if

$$\boxed{w(\tau) = w^*(-\tau)}$$

Since $w(\tau)$ is usually chosen to be real, the constraint becomes $w(\tau) = w(-\tau)$, i.e., the window must be real and even.

(c) **No.** As the original WDF, it cannot distinguish $x(t)$ and $e^{j\theta}x(t)$, since the constant phase cancels in $x(t+\frac{\tau}{2})x^*(t-\frac{\tau}{2})$.

Moreover, the mask causes more information loss. Suppose that $w(\tau) = 0$ for $|\tau| > B$. Taking the IFT along f only recovers $w(\tau)x(t+\frac{\tau}{2})x^*(t-\frac{\tau}{2})$, so with $t_1 = t+\frac{\tau}{2}$ and $t_2 = t-\frac{\tau}{2}$, we only know $x(t_1)x^*(t_2)$ for $|t_1 - t_2| < B$.

Therefore, if $x(t) = s_1(t) + e^{j\theta}s_2(t)$, where s_1 and s_2 are separated by a gap larger than B, all the products between s_1 and s_2 fall outside the mask, and the windowed WDF is the same for every θ.

6. The code (i.e., `recSTFT.py`) is uploaded to NTU Cool, and it is written by the FFT-based method. Using the example input with $B = 1$, the output is shown below.

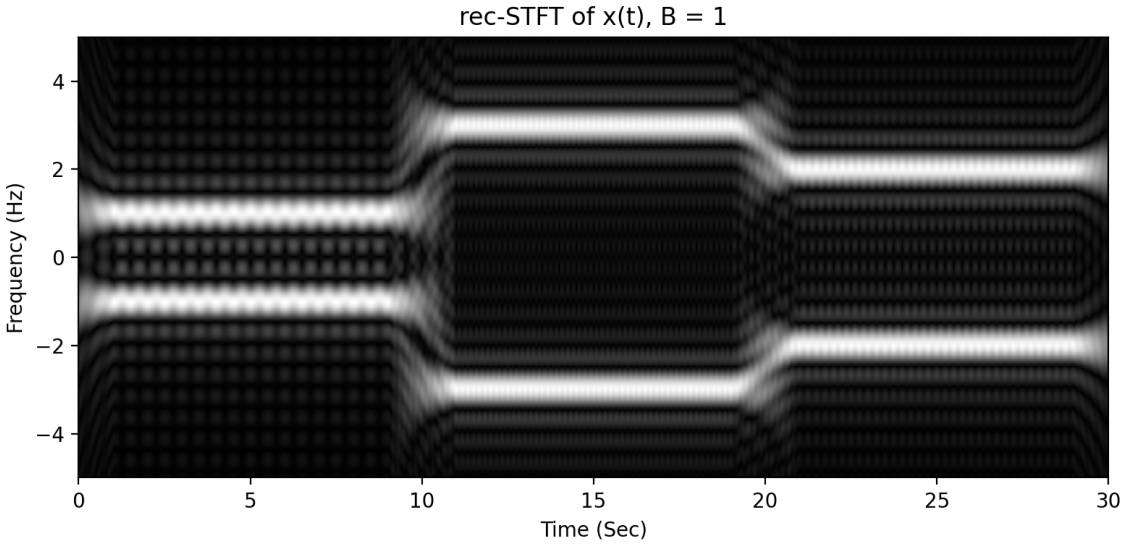


Figure 1: rec-STFT of $x(t)$ with $B = 1$.

The running time is 0.0163 s.

7. (extra)

117

Four Implementation Methods

(1) Direct implementation

Complexity: $TF(2Q+1) \cong 2TFQ$ $\Theta(TFQ)$

假設 t-axis 有 T 個 sampling points, f-axis 有 F 個 sampling points

(2) FFT-based method unbalanced form

Complexity: $\Theta(TN\log N)$ $\Theta(\frac{T}{S}N\log N)$

(3) FFT-based method with recursive formula

Complexity: $\Theta(TF)$

(4) Chirp-Z transform method

Complexity: $\Theta(TN\log N)$

(1) Direct implementation and **(4) Chirp-Z transform method** can also be used in the unbalanced form besides the FFT-based method.

Let $S = \Delta_t/\Delta_\tau$ be an integer, where Δ_τ and Δ_t are the input and output sampling intervals. If the input has T points, the output t-axis has only T/S points. In both methods, each output time is computed independently, so only these T/S points need to be computed:

- Direct implementation: $O\left(\frac{T}{S}FQ\right)$.
- Chirp Z transform: $O\left(\frac{T}{S}N\log_2 N\right)$, since the chirp decomposition still holds with $\Delta_\tau\Delta_f$ in place of $\Delta_t\Delta_f$.

The recursive method gives no saving: each step is derived from the previous one, so when the window moves by S samples, S terms must be removed and S terms added, and the complexity remains $\frac{T}{S}\cdot F\cdot 2S = O(TF)$.